

MATH 200 - Elementary Linear Algebra

MATH 201 - Intermediate Linear Algebra

Section 4.5 Notes

Basis and Dimension

Devotional Thought

Basis of a Vector Space

A nonempty subset of a vector space which is both linearly independent *and* spans the entire space is called a **basis** for the vector space. (The plural of *basis* is *bases*.)

Definition A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ in a vector space V is a basis for V when the conditions below are true.

1. S spans V .
2. S is linearly independent.

Example [Example 1] Show that the set below is a basis for $\mathbb{R}^3$.

$$S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

The basis $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ from the previous example is called the **standard basis** for R^3 . This can be generalized to n -space. That is, the vectors

$$\begin{aligned}\mathbf{e}_1 &= (1, 0, 0, \dots, 0, 0) \\ \mathbf{e}_2 &= (0, 1, 0, \dots, 0, 0) \\ &\vdots \\ \mathbf{e}_n &= (0, 0, 0, \dots, 0, 1)\end{aligned}$$

form the **standard basis** for R^n .

Example [Exercise 2] Write the standard basis for the vector space R^4 .

A basis for a given vector space is *not* unique; there are nonstandard bases for R^n .

Example [Example 3] From examples in section 4.4 we know that the set

$$S = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$$

both spans R^3 and is linearly independent. So S is a basis for R^3 .

Example [Example 2] Show that the set

$$S = \{(1, 2), (1, -2)\}$$

is a basis for $\mathbb{R}^2$.

We can now make explicit what was mentioned in section 4.4 about existence and uniqueness of linear combination representations for elements of a vector space.

Theorem 4.9 Uniqueness of Basis Representation

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a basis for a vector space V , then every vector in V can be written in one and only one way as a linear combination of vectors in S .

Example [Example 6] Let $\mathbf{u} = \{u_1, u_2, u_3\}$ be any vector in R^3 . Show that the equation

$$\mathbf{u} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$$

has a unique solution for the basis

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}.$$

Proof of Theorem 4.9. Let $\mathbf{u}$ be an arbitrary vector in V . Since S is a basis for V , S spans V , and thus we may express $\mathbf{u}$ as a linear combination of the vectors in V :

$$\mathbf{u} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k.$$

This proves the existence of a linear combination representation for $\mathbf{u}$.

To prove the uniqueness of a such a form (that a vector can be represented in only one such way), assume $\mathbf{u}$ has another representation

$$\mathbf{u} = b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \cdots + b_k\mathbf{v}_k.$$

Subtracting the second representation from the first produces

$$\mathbf{u} - \mathbf{u} = (c_1 - b_1)\mathbf{v}_1 + (c_2 - b_2)\mathbf{v}_2 + \cdots + (c_k - b_k)\mathbf{v}_k.$$

S is linearly independent, however, so the only solution to this equation is the trivial solution

$$c_1 - b_1 = 0, \quad c_2 - b_2 = 0, \quad \dots, \quad c_k - b_k = 0$$

which means that $c_i = b_i$ for all $i = 1, 2, \dots, n$, and $\mathbf{u}$ has only one representation for the basis S . ■

That is, spanning gives the existence of a linear combination while linear independence gives the uniqueness.

Comparing Spanning Sets, Linear Independence, and Bases

A basis has two features: A basis S must have enough vectors to span V , but not so many vectors that one of them could be written as a linear combination of the other vectors in S . This can be made more apparent by considering various sizes of spanning and linear independent sets containing and contained in, respectively, the basis.

Theorem 4.10

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a basis for a vector space V , then every set containing more than n vectors in V is linearly dependent.

Exercise 67 from 4.4

A nonempty subset of a finite set of linearly independent vectors is linearly independent.

Exercise 71 from 4.4

Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be a linearly independent set of vectors in a vector space V . Then $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}\}$ cannot span V .

Exercise 72 from 4.4

If V is spanned by $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ and one of these vectors can be written as a linear combination of the other $k - 1$ vectors, then the span of these $k - 1$ vectors is also V .

$$\begin{array}{c}
 V \\
 \cup \\
 S_m \\
 \cup \\
 \vdots \\
 \cup \\
 S_{n+1} \\
 \cup \\
 B \text{ (a basis for } V) \\
 \cup \\
 S_n \\
 \cup \\
 \vdots \\
 \cup \\
 S_1
 \end{array}$$

In Summary: A basis for V is a *minimal* set of spanning vectors for V .

Example [Example 7(a)] Explain why the set $S = \{(1, 2, -1), (1, 1, 0), (2, 3, 0), (5, 0, -1)\}$ must be linearly dependent.

Theorem 4.11 If a vector space V has one basis with n vectors, then every basis for V has n vectors.

Proof. Let $S_1 = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis for V , and let $S_2 = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ be any other basis for V . Theorem 4.10 implies that $m \leq n$, because S_1 is a basis and S_2 is linearly independent. Similarly, $n \leq m$ because S_1 is linearly independent and S_2 is a basis. Consequently, $n = m$. ■

Example [Example 8(a)] Explain why the set $S = \{(3, 2, 1), (7, -1, 4)\}$ is not a basis for R^3 .

On a related note (and demonstrated in Example 6) there is a straightforward computation to determine if a set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for R^n :

For any square matrix A of order n , the column vectors of A form a basis of R^n if and only if $\det(A) \neq 0$.

Example [Example 12] Show that the set of column vectors is a basis for R^5 .

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 3 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 2 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -2 \end{bmatrix} \right\}$$

The Dimension of a Vector Space

By Theorem 4.11, if a vector space V has one basis with n vectors, then every basis for V has n vectors. This number n is the **dimension** of V .

Definition If a vector space V has a basis consisting of n vectors, then the number n is the **dimension** of V , denoted by $\dim(V)$. When V consists of the zero vector alone, the dimension of V is defined as zero.

Most notably, the dimension of $\mathbb{R}^n$ is indeed n .

Intuitively: Dimension is a measure of how many independent variables are needed to uniquely express any point in the space of interest.

Rigorously: Dimension is the number of linearly independent vectors needed to span the vector space.

Equivalently, dimension is the number of vectors needed to write any element of the vector space as a linear combination where the involved coefficients are unique.

Example [Example 9] Find the dimension of each subspace of R^3 .

(a) $W = \{(d, c - d, c) : c \text{ and } d \text{ are real numbers}\}$

(b) $W = \{(2b, b, 0) : b \text{ is a real number}\}$

Remark The set $\{\mathbf{0}\}$ is linearly dependent; this is why $\dim\{\mathbf{0}\} = 0$.

Example [Example 10] Find the dimension of the subspace W of R^4 spanned by

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{(-1, 2, 5, 0), (3, 0, 1, 2), (-5, 4, 9, 2)\}.$$

Usually, to conclude that a set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a basis for a vector space V , you must show that S satisfies two conditions: S spans V and S is linearly independent. If V is known to have a dimension of n , however, then the next theorem tells you that you do not need to check both conditions. Either one will suffice.

Theorem 4.12 Let V be a vector space of dimension n .

1. If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a linearly independent set of vectors in V , then S is a basis for V .
2. If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ spans V , then S is a basis for V .

Example [Exercises 16 and 18] Explain why the set is not a basis for R^3 . What is the dimension of $\text{span}(S)$?

- $S = \{(2, 1, -2), (-2, -1, 2), (4, 2, -4)\}$

- $S = \{(1, 1, 2), (0, 2, 1)\}$

Example [Exercises 8, 10, 12] Explain why the set is not a basis for $\mathbb{R}^2$. What is the dimension of $\text{span}(S)$?

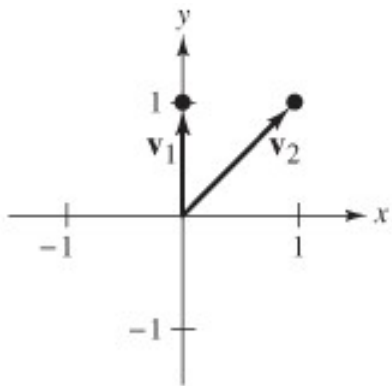
- $S = \{(2, 3), (6, 9)\}$

- $S = \{(5, -7)\}$

- $S = \{(-1, 2), (1, -2), (2, 4)\}$

More Exercises (for extra practice as time allows)

Example [Exercise 38] Determine whether the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis for $\mathbb{R}^2$.



Example [Exercise 68] Find all subsets of the set

$$S = \{(1, 3, -2), (-4, 1, 1), (-2, 7, -3), (2, 1, 1)\}$$

that form a basis for R^3 .

Example [Exercises 72 and 74]

(a) Give a geometric description of, (b) find a basis for, and (c) find the dimension of the given subspace W of $\mathbb{R}^n$ for the given n .

- $W = \{(0, t) : t \text{ is a real number}\}, n = 2$

- $W = \{(2s - t, s, t) : s \text{ and } t \text{ are real numbers}\}, n = 3$

Example [Exercise 76]

Find (a) a basis for and (b) the dimension of the subspace W of R^4 .

$$W = \{(5t, -3t, t, t) : t \text{ is a real number}\}$$